

Build Prompt — paste this into Claude Code

I want to build "DenjiPet," a Python desktop pet app (Windows-first), inspired by apps like Comnyang and classic Shimeji-style desktop companions. Skin/theme is Chainsaw Man's Denji, but treat all character art as a pluggable asset folder — don't generate or hardcode any character artwork yourself, just wire up the loading logic so I can drop in my own sprite PNGs later.

Stack

- Python 3.11+
- PySide6 for the GUI
- PyInstaller for packaging (later step, not now)

Build this incrementally, confirming each step runs before moving to the next:

Step 1 — Base window

Create a frameless, transparent, always-on-top PySide6 QWidget that:

- Has no taskbar icon
- Displays a single placeholder image (use a simple colored circle or fallback icon if no sprite asset exists yet at `assets/idle/frame_0.png`)
- Is draggable anywhere on screen via left-click-drag
- Right-click opens a context menu with "Quit"

Step 2 — Sprite animation loop

- Build `animation_controller.py`: loads all PNGs from a given folder (e.g. `assets/idle/`), cycles frames on a QTimer at a configurable FPS (default 6 fps), loops indefinitely
- Wire it into the base window so the idle animation plays automatically

Step 3 — State machine + Pomodoro engine

- Build `pomodoro_engine.py` as a simple state machine: IDLE → WORKING → BREAK → (repeat), with a LONG_BREAK every 4th cycle
- Default durations: 25 min work, 5 min break, 15 min long break (all configurable via constructor args)
- Emit a Qt signal on every state transition
- Connect that signal to `animation_controller` so it swaps the active sprite folder based on state (idle/working/break/tired mapped from state)

Step 4 — System tray

- Build `tray_menu.py` using QSystemTrayIcon

- Menu items: Start/Pause Pomodoro, Reset, Settings (placeholder), Quit
- Tray icon click toggles pause/resume of the Pomodoro engine

Step 5 — Health nudges

- Build `nudge_manager.py`: a separate QTimer firing every N minutes (default 20), picks a random reminder from a hardcoded list ("Drink water", "Stretch your back", "Blink and look away from the screen")
- Displays it as a small speech-bubble-style popup anchored near the pet window, auto-dismisses after ~5 seconds

Step 6 — Settings persistence

- Build `settings.py`: loads/saves a JSON config at `%APPDATA%/DenjiPet/config.json` on Windows (fallback to local dir on other OS for dev testing)
- Persist: work/break durations, nudge interval, sprite folder path
- Load these on startup instead of hardcoded defaults

Step 7 — Packaging (only after everything above works)

- Write a PyInstaller spec/command to bundle this into a single `.exe`
- Make sure the `assets/` folder is included as a data directory

Constraints

- No character artwork generation — sprite folders may be empty/placeholder during dev; code should handle missing assets gracefully (fallback shape)
- Comment the code reasonably since I'm a data science undergrad newer to desktop/GUI Python, not a beginner to Python itself
- Keep each file focused/single-responsibility per the folder structure below

Folder structure to follow

```
DenjiPet/  
├─ main.py  
├─ pet_window.py  
├─ animation_controller.py  
├─ pomodoro_engine.py  
├─ nudge_manager.py  
├─ tray_menu.py  
├─ settings.py  
├─ assets/  
│   ├── idle/  
│   ├── working/  
│   ├── break/  
│   ├── tired/  
│   └── celebrate/  
└─ requirements.txt
```

Start with Step 1 and show me the running window before moving on.